

M3 — Narrowphase (contact manifolds) ✓

All three shape pairings done, bit-for-bit vs the real original Nape AS3: **M3a circle-circle** ✓ · **M3b circle-polygon** ✓ · **M3c polygon-polygon** ✓.

The game uses all three (a ball is a circle; terrain and crates are polygons; the crate is a *dynamic* polygon, so poly-poly is real).

Tested vs the real original Nape AS3. The manifold is observed via the public arbiter API after one `space.step ( CollisionArbiter.normal , Contact.penetration )` — no solver needed, since narrowphase runs pre-solve.

Scope finding

Nape's narrowphase is **one ~1100-line function**, `contactCollide ( ZPP_Collide.as:176 )`, covering all three shape pairings:

- **circle-circle** (lines 1088–1300) — self-contained, **done**;
- **circle-polygon** (~950–1080) — closest-feature;
- **polygon-polygon** (219–~950) — SAT over edge normals + clipping. The bulk.

Key discovery: Nape normalizes with a **Quake fast-inverse-sqrt** (float32 bit-hack `1597463007 - (i>>1)` + one Newton step), *not* `Math.sqrt`. So normals are **not unit length** (e.g. 0.99977) and distances carry the approximation error. The replica reproduces this bit-for-bit via a `Float32Array / Int32Array` round-trip (`fastInvSqrt` in `nape-core.ts`). This matters everywhere Nape normalizes — including poly-poly — so it's a prerequisite the rest of M3 builds on.

M3a — circle-circle ✓

Replica: `circleCircleManifold(ha, hb)` — radii sum, centre delta, fast-inv-sqrt normal, penetration. Bodies passed in the original's `(body1, body2)` order (the normal sign depends on it; the harness traces that order).

Gate (`m3-circle.test.ts`): two overlapping circles (r20, centres 30 apart):

- `normal = (-0.99977, -0)` — bit-for-bit, fast-inv-sqrt and all;
- `penetration = 9.99320` — bit-for-bit.

✓ manifold normal + penetration match the original bit-for-bit (incl. fast-inv-sqrt)

M3b — circle-polygon ✓

`circlePolyManifold ( nape-core.ts )`: deepest-edge loop (`sep = gnorm·c - gproj - r`), then a region test → edge-face contact (exact) or vertex contact (fast-inv-sqrt). Edge normals computed Nape's way (edge dir rotated 90°, normalized with `Math.sqrt` — bit-exact across AVM2/V8). Gate (`m3b-circlepoly.test.ts`): EDGE (0,-1) pen 2.0; VERTEX (-0.35870,-0.93263) pen 1.06092 — both bit-for-bit.

M3c — polygon-polygon

`polyPolyManifold` (`nape-core.ts`): SAT over both polys' edge normals → pick reference/incident edge → clip the incident edge to the reference side bounds → up to 2 contacts (point + penetration). Gate (`m3c-polypoly.test.ts`): two overlapping boxes → normal `(0,1)` , two contacts pen 8.0 at `(180,192)` / `(220,192)` , bit-for-bit (contact set matched order-independently; ordering/ids handled in M4).

Deferred to M4

Contact **ids** (`hash`) and ordering for warm-starting; contact `position` getter nuances; how `param4` / `rev` are derived from arbiter shape order (the manifold tests pin `param1=body1` directly).